Central Tendency: Mean, Median, Mode

Measure	Meaning	Effect of outliers	Example use	Code
Mean	Sum of values ÷ number of values	Strongly affected	Average daily shipments (600 ÷ 5 = 120)	`dataset["ssc_p"].mean()`
Median	Middle value in sorted data	Generally less affected	Typical salary (AED 3,000 vs. mean 5,200)	`dataset["ssc_p"].median()`
Mode	Most frequent value	Generally not affected	Best-selling size (Medium, 50 sales)	`dataset["ssc_p"].mode()[0]`

To fill a results table cell:

```
descriptive.loc["Mean", columnName] = dataset[columnName].mean()
```

Results

Category	Mean	Median	Mode
SSC	67.30	67.00	62.00
HSC	66.33	65.00	63.00
Degree	66.37	66.00	65.00
Entrance test	72.10	71.00	60.00
MBA	62.28	62.00	56.70
Salary	***288,655.41***	***265,000***	***300,000***

The Story of 108 Students

Across their journey from 10th standard to MBA, the students averaged 67.30% in SSC and 66.33% in HSC. Their degree average stayed almost unchanged at 66.37%. The entrance test was their strongest stage at 72.10%, while the MBA average was the lowest at 62.28%. Their salary mean (288,655.41) is above the median (265,000), suggesting a few high salaries pull the average up.

Conclusion: The mean shows the average, the median shows the middle, and the mode shows the most frequent value. Together, they describe the students' academic performance and salary distribution.